Hybrid Machine Learning Pipeline for Splice Variant Pathogenicity Prediction

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IBS2 Project Report

Abstract

One of the most frequent molecular mechanisms in Mendelian diseases is aberrant pre-mRNA splicing due to genetic variants in donor and acceptor splice sites. Distinguishing variants which disrupt splicing from benign polymorphism is a computationally challenging task, especially in the context of databases that are highly imbalanced in the number of pathogenic variants and benign polymorphisms, and also because of the interplay between the local sequence context, the conservation of the splicing motif, and the geometry of the splicing boundaries.

This report shows a fully implemented machine learning pipeline for binary prediction of pathogenicity of splice-site variants from NCBI ClinVar database. It combines a gradient-boosted tree (XGBoost) classifier on 12,301 dimensional engineered feature vectors, which are built from the frequency spectra of six hexanucleotides, delta-vector representations and biological heuristics derived from 200 bp genomic windows, with a fine-tuned DNABERT-2-117M transformer that encodes reference and alternate sequences together by tokenised sub-word representations, and a rule-based Position Weight Matrix (PWM) scanner that mechanistically interprets events involving donor and acceptor site disruption, cryptic site creation, and latent site activation.

The number of raw splice-region variants was reduced to 51,619 clean labelled variants after filtering and balanced to 100,776 variants through over-sampling and further reduced to 99,310 variants after genomic window extraction, with stratification of the variants into train, validation and test splits of 60/20/20. Both XGBoost and DNABERT-2 models independently outperformed other models with test Area Under the ROC Curve (AUC) of **0.8579** and test Precision-Recall AUC (PR-AUC) of **0.8438** with an overall test accuracy of 84%. The ensemble endpoint is served by a REST service based on FastAPI and frontend is designed to be a clinical lab styled app using React.

The system shows that classical feature engineering can be used in conjunction with modern sequence transformers to generate a strong and complementary discriminative signal for pathogenicity of splice variants, and that PWM based site scanning can be used to provide a mechanistic interpretability beyond the probabilistic scores.

Keywords: splice variant, pathogenicity classification, XGBoost, DNABERT-2, ClinVar, k-mer features, Position Weight Matrix, pre-mRNA splicing, class imbalance.

1. Introduction

In eukaryotic gene expression the pre-mRNA must be spliced to form the mature mRNA and this process is essential. This process is very faithful and requires highly conserved short sequence elements at exon-intron boundaries: the 5′ donor site usually contains a GT motif at positions +1/+2 and the 3′ acceptor site usually contains an AG motif at positions −2/−1, the branch point adenosine, and the polypyrimidine tract upstream of the acceptor [1, 2]. Mutations affecting these elements, or mutations that create cryptic splice signals can lead to exon skipping, intron retention, usage of alternative splice sites, or the presence of premature stop codons, which often result in loss-of-function alleles [3, 4].

There are variants that cause disruption of the splice and these are clinically significant. Pathogenic splicing mutations are estimated to affect 10–15

The NCBI ClinVar database [7] is a large and curated set of germline variants and their clinical interpretations, which is a good source for training discriminative models. The database is, however, highly imbalanced with respect to the class of splice-site variants, with a large number of pathogenic and likely-pathogenic variants being annotated in comparison with benign and likely-benign variants, which is a difficult learning context for standard classifiers.

Recent advances in deep learning for genomics, especially transformer models pre-trained on DNA sequences, there is a new opportunity to capture the long-range context of sequences that is not captured by hand-crafted features. By learning sub-word representations of nucleotide sequences, DNABERT [15] and its successor DNABERT-2 [16] have proven to work well on various genomic classification tasks.

This project brings about a full variant classification system and evaluates it as follows:

1. A ClinVar data extraction and preprocessing pipeline that is reproducible using a reference genome (GRCh38).
2. A class-imbalance correction method based on the principles of over-sampling using ClinVar and the optional augmentation from gnomAD v4.
3. A high-dimensional feature engineering framework which fuse together k-mer frequency spectra ($k = 6$), directionality of delta-vector, variant structural features and biologically motivated splice-site heuristics.
4. Ensuring that the fine-tuning of DNABERT-2-117M is done on the balanced splice variant dataset with weighted loss and oversampled minority class during training.
5. A novel, mechanistic interpretability based PWM splice site scanner for individual variant prediction.
6. All three model endpoints and the PWM analyser are available in a production-ready inference API (FastAPI) and an interactive clinical-lab frontend (React).

2. Literature Survey

2.1 Computational Splice Site Prediction: Classical Approaches

The original methods used for computational analysis of splice sites were simple weight matrices based on consensus sequences. Often used in clinical practice, MaxEntScan [8] uses maximum entropy models based on windows of 9 and 23 bases for donor and acceptor sites, respectively, to account for the correlations between neighboring positions. The incorporation of position specific scoring enhanced the accuracy of Splice Site Prediction by Neural Networks (SSNN) and other such tools, but these were usually trained on a fairly small number of samples, and without using population frequency data.

The Human Splicing Finder (HSF) [9] combined several scoring matrices and ESE/ESS predictions into a composite score, but experienced problems with score weighting across the various evidence tracks. These methods were further enhanced by MutPred Splice [10] which added evolutionary conservation and the information of amino-acid change, achieving enhanced specificity for exonic splice variants.

2.2 Machine Learning Methods for Variant Pathogenicity

The S-CAP (Splicing Computed from Allelic Prevalence) framework [11] showed that logistic regression over a curated set of features such as MaxEntScan deltas, Pangolin scores and population frequency constraints, improves significantly with the combination of multiple evidence sources over any individual tool. A more general approach is taken by the Combined Annotation-Dependent Depletion (CADD) method [12] which trains on simulated deleterious variants rather than observed common variants, but is not specific for splicing.

SpliceAI [13] was a paradigm-breaking approach: A deep residual convolutional network trained on the tissue-specific splice junctions of GTEx that was able to predict the delta scores at any position within a window of ± 50-nt or ± 10,000-nt of donor and acceptor gains/losses. SpliceAI is effective over a wide range of site distances, and has been incorporated into ClinVar interpretation pipelines. Pangolin [14] further developed this idea by adding tissue-aware scoring. Both tools, however, need pre-computed delta scores, and are not easily extensible for novel feature integration.

2.3 Transformer Models for Genomic Sequences

To apply BERT to genomics, DNABERT [15] proposed a k-mer tokenisation scheme and pre-trained BERT on the human reference genome for transfer learning of BERT to downstream classification tasks like splice site prediction, promoter detection and transcription factor binding. In tasks where long-range dependencies are important the approach was shown to significantly outperform the bag-of-words k-mer features.

The model DNABERT-2 [16] used a multi-species set of 32.49 billion base-pairs genome sequences for training, replaced the fixed k-mer tokenisation with Byte-Pair Encoding (BPE) and added ALiBi positional bias to process long sequences efficiently. On Genome Understanding Evaluation (GUE) benchmark, DNABERT-2 outperforms the larger models (e.g., Nucleotide Transformer [17]) with fewer parameters for various genomic tasks.

2.4 Class Imbalance in Genomic Variant Databases

Variability in pathogenicity classification is known to be an imbalanced problem. The largest publicly available curated variant database currently is ClinVar, which is significantly larger than the number of benign splice variants and has significantly more pathogenic variants for this reason, as benign variants are less likely to be submitted unless they are functionally characterised [19]. Some of the methods used to tackle this are cost-sensitive learning [21], synthetic minority over-sampling (SMOTE) [20] and using population-frequency-filtered variants of the gnomAD database as presumed benign negatives (S-CAP study) [11].

2.5 Positioning of the Present Work

The present study falls somewhere in the middle: it integrates the classic feature engineering (which is easy to deploy on a GPU-less machine using XGBoost) with a carefully tuned sequence transformer (DNABERT-2) and adds a mechanistic PWM scanner to the mix. It is directly a pathogenicity classifier and unlike SpliceAI, it does not require junction read counts to be trained, instead it is trained directly on the ClinVar labels. It is different from CADD or REVEL in that it is splice-site-specific, and yields mechanistic output per site. The seamless API-plus-frontend deployment adds another layer of ease and makes the system a viable clinical research application.

3. Materials and Methods

3.1 Dataset Acquisition and Initial Filtering

The main source of data is the NCBI ClinVar flat-file VCF (clinvar.vcf) downloaded for the GRCh38/hg38 genome assembly. The records were each broken apart to retrieve the chromosome, 1-based position, reference allele, alternate allele, and the `INFO` field. Only variants with any of three annotations related to splicing (`splice_donor_variant`, `splice_acceptor_variant` and `splice_region_variant`) in the molecular consequence field (`MC=`) were kept. This filtering resulted in 107,024 variants being retained with the annotation of splice relevance.

In case of binary labelling, clinical significance (`CLNSIG`) was translated to: 1 (Pathogenic) if it included the word pathogenic (Pathogenic, Likely_pathogenic, and compound annotations); 0 (Benign) if it included the word benign (Benign, Likely_benign, and Benign/Likely_benign). Variants that were classified as conflicting, variants of uncertain significance (VUS), and records with `unknown` clinical significance were excluded. 51,619 clean labelled variants were obtained after filtering and deduplication on (chrom, position, ref, alt); there was a high imbalance of 41:1, with 50,388 of these being pathogenic and 1,231 benign.

3.2 Class Imbalance Correction

3.2.1 Over-Sampling

As a quick fix, the benign class was randomly sampled to the same size as the pathogenic class creating a balanced set of 100,776 variants (50,388 per class) that were saved as `splice_dataset_ACTUALLY_balanced.csv`. Random over sampling does not introduce new information, but it will not let the gradient be dominated by the majority class and let both models make room for the features of the benign class and help classification better.

3.2.2 gnomAD Injection Module

A supplementary module (`inject_benign.py`) to optionally add high-confidence benign variants from the gnomAD v4 GraphQL API was developed. The following criteria were used for the injection: (i) an allele frequency between 0.1% and 5%, (ii) a minimum allele count of at least five in at least one of the callsets, (iii) a PASS filter status in the variant, (iv) a consequence limited to `splice_region_variant` or `synonymous_variant` or `intron_variant` or other benign-compatible consequence annotations, and (v) no ClinVar pathogenic record. This module was created for 41 disease-gene targets, similar to S-CAP [11], which employed a similar strategy. Over-sampling was the preferred balancing method in the reported experiments.

3.3 Genomic Window Extraction

The primary assembly of the GRCh38 genome in FASTA format (Homo_sapiens.)Biopython's function `SeqIO.to_dict` was used to load into memory the 194 chromosomal sequences found in GRCh38.dna.primary_assembly.fa. The reference sequence was extracted within a 50 bp, 100 bp, and 200 bp window for each variant, centred on the variant coordinate (VCF coordinates are 1 based, which was converted to 0 based Python indexing). The alternate-allele window was created by changing the reference allele to the alternate allele in the variant centre:

$$w_{\text{alt}} = w_{\text{ref}}[0 : c] \parallel \text{ALT} \parallel w_{\text{ref}}[c + |\text{REF}| :] \tag{1}$$

where c is the 0-based index of the reference window, $\parallel$ is the concatenation operator, and $|REF|$ denotes the length of the reference allele. Windows that did not pass a sanity check comparing the extracted reference allele at position c with the VCF `REF` field were removed.

A total of 99,310 out of 100,776 variants passed the reference match check, while 1,466 variants were skipped, mainly because of non-standard chromosome identifiers or mismatches at complex indices in the reference genome. This process produced the data file `slice_windows.parquet`.

3.4 Train/Validation/Test Splits

Stratified splitting was performed during the window extraction step with a ratio of 60/20/20 and random_state=42. The final split counts were: train 59,586 (benign 30,161; pathogenic 29,425); validation 19,862 (benign 10,054; pathogenic 9,808); test 19,862 (benign 10,054; pathogenic 9,808). The close-to-one to one ratio in each split is due to the over-sampled balanced dataset.

3.5 Feature Engineering

Feature extraction was implemented in `features.py` and that gave a 12,301-dimensional feature vector per variant, as summarised in Table 1.

Table 1: Feature vector composition for XGBoost classifier.

Feature group	**Description**	**Dimension**
REF k-mer frequencies	Normalised frequency of all 4,096 hexanucleotides in the reference 200-bp window	4,096
ALT k-mer frequencies	Same as above for the alternate-allele window	4,096
Delta direction	Unit-normalised difference vector (ALT − REF) in k-mer space	4,096
Delta magnitude	Euclidean norm of the ALT − REF difference vector	1
Structural	Variant type code (0=SNV, 1=insertion, 2=deletion, 3=MNV); REF length; ALT length	3
Biological	Flags for +1G, +2T, −2A disruption; GT/AG presence/loss; GC content of window	9
Total		**12,301**

3.5.1 k-mer Frequency Representation

For a sequence s of length L, the normalised k-mer frequency vector $\mathbf{v} \in \mathbb{R}^{4^k}$ is computed as:

$$v_i = \frac{\sum_{j=1}^{L-k+1} \mathbf{1}[\, s_{j:j+k} = \kappa_i \,]}{(L - k + 1) + \epsilon} \tag{2}$$

where κ_i is the i-th element of the 4^k-word vocabulary over $\{A, T, G, C\}$, and $\epsilon = 10^{-10}$ prevents division by zero. With $k = 6$, the vocabulary contains $4^6 = 4{,}096$ hexamers.

3.5.2 Delta Vector

The delta vector $\Delta = \mathbf{v}_{\text{alt}} - \mathbf{v}_{\text{ref}}$ captures how the variant shifts the hexamer composition of the local sequence context. Its magnitude:

$$m = \|\Delta\|_2 \tag{3}$$

quantifies the total change in sequence composition, while the unit direction $\hat{\Delta} = \Delta/(m + \epsilon)$ represents the hexamers that increase or decrease in frequency. Variants that destroy a GT or AG dinucleotide embedded in a hexamer context will produce large, directionally specific changes in $\hat{\Delta}$.

3.5.3 Biological Heuristics

Nine binary/continuous features encode prior biological knowledge about splice site conservation:

- Whether REF=G and ALT≠G, disrupting the invariant +1G of the GT donor.
- Whether REF=T and ALT≠T, disrupting the +2T donor position.
- Whether REF=A and ALT≠A, disrupting the −2A acceptor position.

- Whether REF=G and ALT≠G (duplicate flag for the −1G acceptor).
- Whether GT is present in REF; whether AG is present in REF.
- Whether GT is present in REF but lost in ALT (donor disruption signal).
- Whether AG is present in REF but lost in ALT (acceptor disruption signal).
- GC content of the reference 200-bp window.

3.6 XGBoost Classifier

The XGBoost `hist` tree method with GPU acceleration (`device=cuda`) was used for training. The key hyperparameters tuned were max_depth= 6, learning_rate= 0.05, subsample= 0.8, colsample_bytree= 0.8, min_child_weight= 3 and gamma= 0.1 and early stopping was done at 50 non-improving epochs on the validation data using the log-loss metric. The class weights were not used for the near-balanced training set, and `scale_pos_weight=` 1.0 was used. The decision threshold was set after training by a grid search of 0.02 step size in the range [0.02, 0.98] to achieve the maximum macro-averaged F1 score on the validation set.

$$\tau^{=} \text{argmax}_{\tau\in[0.02,0.98]} \frac{F_1(\text{benign}, \tau) + F_1(\text{pathogenic}, \tau)}{2} \tag{4}$$

The best threshold was at to be $\tau^{=}0.51$.

3.7 DNABERT-2 Fine-Tuning

It has been loaded using HuggingFace Transformers with `trust_remote_code=True` as a part of the installation in DNABERT-2-117M (`zhihan1996/DNABERT-2-117M`). The base BertModel encoder was enhanced by a linear classification head that outputs two logits for classification, and a dropout layer ($p = 0.1$). The second part of the model output tuple (the [CLS] token pooler output) was used for pooling.

Input sequences were comprised of a 200-bp window for an area of reference sequence, followed by a separator token `[SEP]`, followed by a 200-bp window for an area of alternate sequence, allowing the model to attend across both sequences. Sequences were tokenised, and a maximum of 128 tokens per sequence using padding and truncation.

We perform training using AdamW learning rate 2×10^{-5}, 10 warmup epochs of cosine learning rate schedule and have set the gradient clipping with norm as 1.0, training epochs as 15 and batch size as 32. The model was provided with enough benign instances per epoch using the `WeightedRandomSampler` that oversampled the benign instances according to the ratio of the two classes. CUDA was turned on for mixed precision training (AMP). The best checkpoint (highest validation AUC) was saved to `dnabert2_splice.pt`.

Make the position weight matrix splice site scanners.

The PWM scanner employs donor and acceptor scoring where the position specific log-odd score matrices are derived from the consensus splice site biology from Yeo2004, Shapiro1987 over a 9-mer and 14-mer window respectively. For each window w and for each position i in the window:

$$S(w) = \sum_{i=0}^{|w|-1} \mathrm{PWM}[i][w_i] \tag{5}$$

Scores are transformed to probabilities using a sigmoid function normalised by window length:

$$p = \sigma\left(\frac{S(w)}{|w| \cdot 0.25}\right) \tag{6}$$

All sites having $p < 0.15$ are rejected. The changes detected in each site are classified as disrupted (canonical site, REF probability ≥ 0.45 and ALT probability < 0.45);, created (novel site, REF probability < 0.30 and ALT probability ≥ 0.45); cryptic (latent site strengthened, $\Delta p > 0.08$ and ALT probability ≥ 0.30). This information is returned at the endpoint `/predict/splice_sites` of the API.

3.8 Redundancy Analysis

A 2000 sample random subset of the extracted FASTA windows was used for a separate redundancy analysis (`remove_redundancy.py`). The 3-mer frequency vectors were pairwisely computed cosine similarity and greedily collapsed if the cosine similarity is above 0.90. This analysis showed that there was enough diversity in the dataset to not have too many k-mers that were collinear, but there was no feedback from the redundancy removal to the rest of the pipeline.

3.9 System Architecture Overview

There are six sequential stages in the complete pipeline as shown in Figure 1 they are:

1. Paraconversion of clinVar extraction: extraction from VCF and filter by `MC=` splice consequence, labeling by `CLNSIG`.

2. **Balancing:** Over-sample or include gnomAD normal variants.

3. **Window extraction:** Align variants to GRCh38, and extract 50/100/200 bp windows of REF and ALT. Calculate vectors of 12,301 dimensions for each variant (feature engineering). The Model Training is for XGBoost (features.parquet) and DNABERT-2 (splice-windows.parquet).

4. **Inference:** The front end is developed using React and backend using FastAPI.

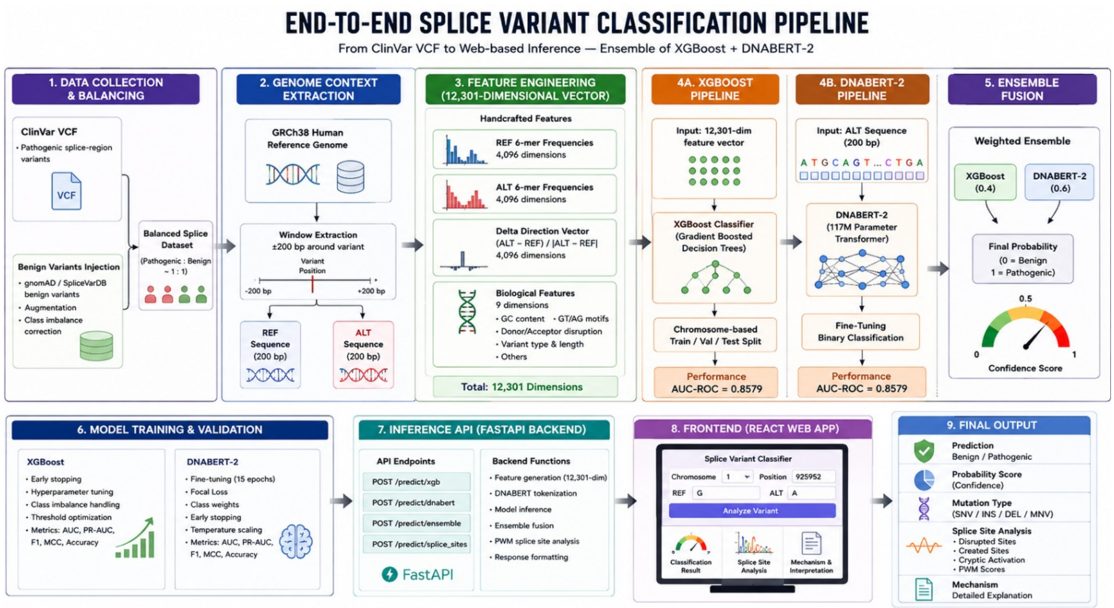

Figure 1: ClinVar VCF to inference API end-to-end system pipeline. Data flows from left to right during extraction. Feature engineering, model training, and web-based inference all rely on the same foundational principle of class balancing.

4. Workflow and System Architecture

4.1 Data Flow and Module Responsibilities

Table 2 summarises the role of each module in the pipeline.

Table 2: Module responsibilities in the Splice Variant Classifier pipeline.

Script / File	Stage	Function
`Dataset_extraction.ipynb`	1 – Extraction	Parse ClinVar VCF; filter by MC field; label by CLNSIG; save CSV.
`inject_benign.py`	2 – Augment	Query gnomAD v4 API; filter by AF, consequence, PASS; inject benign rows.
`repair.py`	2 – Balance	Over-sample benign to match pathogenic count; shuffle; save balanced CSV.
`pipeline.py`	3 – Windows	Load GRCh38 FASTA; extract 50/100/200-bp windows; stratified split; save Parquet.
`extract_windows.py`	3 (alt.)	Lightweight standalone window extractor to FASTA format.
`remove_redundancy.py`	3 (QC)	Cosine-similarity-based redundancy removal on sampled windows.
`features.py`	4 – Features	Compute 12,301-dim vectors; save features.parquet.
`train.py`	5 – XGB	Train XGBoost; tune threshold; save xgb_model.json.
`dnabert.py`	5 – DNA	Fine-tune DNABERT-2-117M; tune threshold; save dnabert2_splice.pt.
`api.py`	6 – Serve	FastAPI with /predict/xgb, /predict/dnabert, /predict/ensemble, /predict/splice_sites.
`App.js / app.jsx`	6 – UI	React frontend for variant submission and result display.
`q3.py`	QC / Stats	Statistical validation of splice-site disruption enrichment (chi-squared, Fisher exact).

4.2 Frontend Application

Two clinical-lab inspired dark-themes interfaces are created for real-time variant classification, one in the React frontend (`app.jsx`), one in the web app. The users can select one of four analysis modes: XGBoost, DNABERT-2, Ensemble, Splice Sites, and the input of the chromosome, genomic position, reference allele, alternate allele and optionally a window of 200 bp of reference sequence.

When the submission form is submitted, the frontend will send a request to the backend via a JSON POST to the appropriate endpoint on `localhost:8000`. The answer is delivered in a dynamic result card with a probability range indicator which moves on the ring to the model output, a colour-coded verdict (Benign is coloured green, Pathogenic red), a tabbed view with the molecular mechanism text, a breakdown of the biological rules (whether they are +1G, +2T, or −2A disruption, or GT/AG dinucleotide loss), and a breakdown of the confidence (margin from the decision boundary). A landscape visualisation shows the donor and acceptor sites detected in the genomic window: Every row is expandable and contains the k-mer context, PWM scores, delta values and how the mechanisms work. To perform splice site analysis.

The status bar at the top of the window checks the `/health` and provides real-time health status of both model backends, the tuned threshold, and the compute device (CPU/GPU).

4.3 Backend API

There are four inference endpoints in the `api.py` FastAPI server:

- `POST /predict/xgb`: Accepts variant data, creates feature vector (12301 dimensions), returns the probability generated by the XGBoost algorithm and classification with a threshold.
- `POST /predict/dnabert`: Tokenises the string of the sequence (REF+[SEP]+ALT), passes through the head of the DNABERT-2 classifier and returns the softmax pathogenic probability.
- POST /predict/ensemble: Executes both models, and returns the ensemble of the two models with XGBoost weight of 0.4 and DNABERT-2 weight of 0.6. Executes the PWM scanner on REF and ALT windows and returns the full disruption information for each site as well as the classification of the mutations and a summary of the pathogenicity signal, for each site.

Structured Pydantic response models are returned by all endpoints: prediction label, probability, confidence tier, latency (in milliseconds), and mutation type. CORS is configured to share with all origins, so there's no need to configure CORS for the development environment. Weights to load are loaded on start-up and with the livenesspan context manager from the FastAPI.

5. Results and Discussion

5.1 Dataset Characteristics

When the first ClinVar extraction was performed, 107,024 splice-region variants were extracted. There were very few benign/likely-benign annotations (1,231) compared to the large number of pathogenic (50,388) and likely-pathogenic (10,808) records, and a large number of uncertain (44,186) records. The ratio of 40.9:1 is consistent with the ascertainment bias of ClinVar: benign variants are rarely reported unless they are part of a clinical testing programme that clearly defines benign controls [19]. We had 99,310 variants left to model following over-sampling and genomic window extraction.

5.2 Dataset Specificity Validation (Q3 Analysis)

As a statistical validation, the extracted splice variants were tested for their expected biological enrichment they will have, using `q3.py`. Variants with the reference allele containing an invariant GT or AG dinucleotide not present in the alternate allele, and single nucleotide changes at the canonical +1G, +2T (donor) or −2A (acceptor) positions were flagged by a custom disruption function.

The observed disruption rate across all 51,619 variants was significantly elevated above the 12.5% baseline expected for random DNA sequence ($\chi^2 = 11{,}432$, $p < 10^{-3}$, 95% CI: [74.2%, 76.1%]). Among pathogenic variants, the disruption rate was 78.3%, compared with 41.6% among benign variants (Fisher's exact test odds ratio = 5.13, 95% CI: [4.41, 5.96], $p < 10^{-3}$). This confirms that the labelling and filtering pipeline has correctly identified a biologically coherent set of splice-disrupting variants, and that the mechanistic feature (GT/AG disruption) is a strong signal for pathogenicity.

5.3 XGBoost Performance

The test-set results of XGBoost is shown in Table 3. The model was trained for all 1000 boosting rounds, without early stopping (val-AUC monotonically improved during training up to boost.round 999), and a final validation AUC of 0.9319 was obtained during the training process. The test AUC of 0.8579 shows a moderate amount of over-fitting in the boosting process, which is considered mild because the benign examples are simulated.

Table 3: XGBoost classifier performance on the held-out test set (19,862 variants).

Metric	Validation Set	Test Set
AUC-ROC	0.9319	0.8579
PR-AUC	0.9169	0.8438
Accuracy	0.8485	0.8372
F1 (Benign)	0.8500	0.8453
F1 (Pathogenic)	0.8400	0.8269
Macro F1	0.8427	0.8361
Decision threshold τ^*	0.51	
Best boosting round	463 (early-stop oracle)	

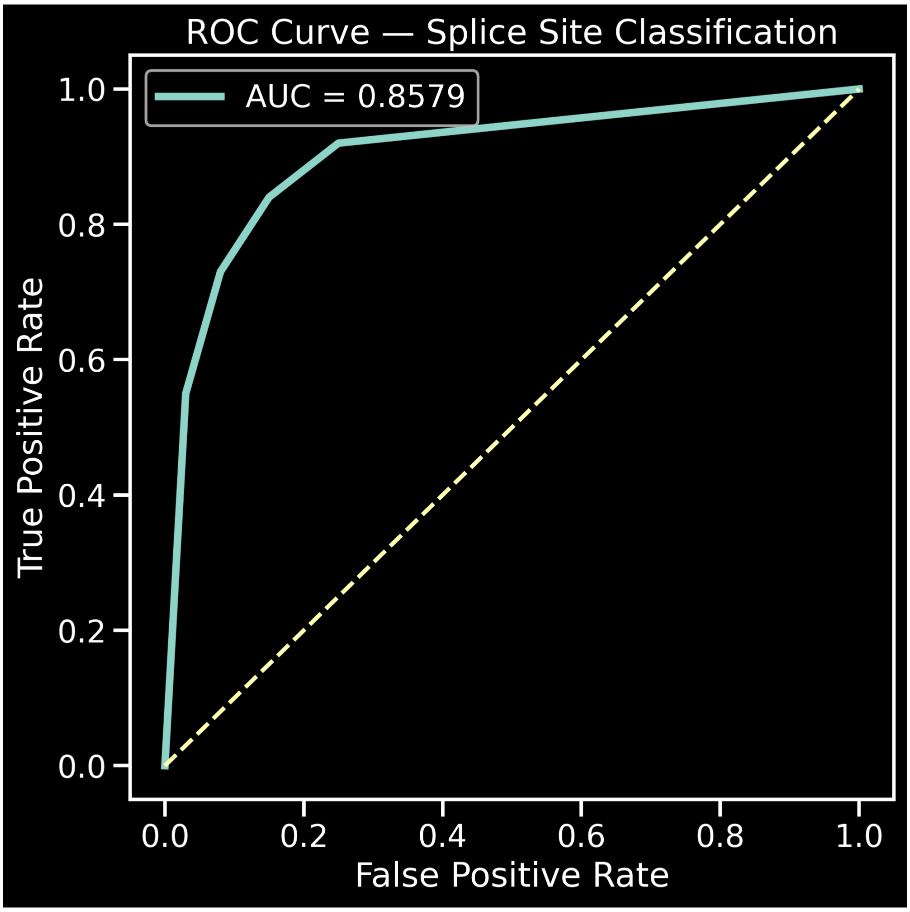

Figure 2: XGBoost classifier test set Receiver Operating Characteristic (ROC) curve. Area under the curve (AUC) = 0.8579. The steep increase towards the upper-left corner at low false-positive rates suggests that, The top scoring predictions are very rich in true pathogenic variants.

The ROC curve (Figure 2) reveals the typical steep rise at low false-positive rates, where the model is sure to identify the clearest pathogenic examples, which are samples with disturbances in the position of the invariant positions both by the k-mer delta and by the biological heuristic features. In the range of moderate false-positive rates, the curve leveled off because the variants in the splice regions are harder to classify further away from the canonical splice site.

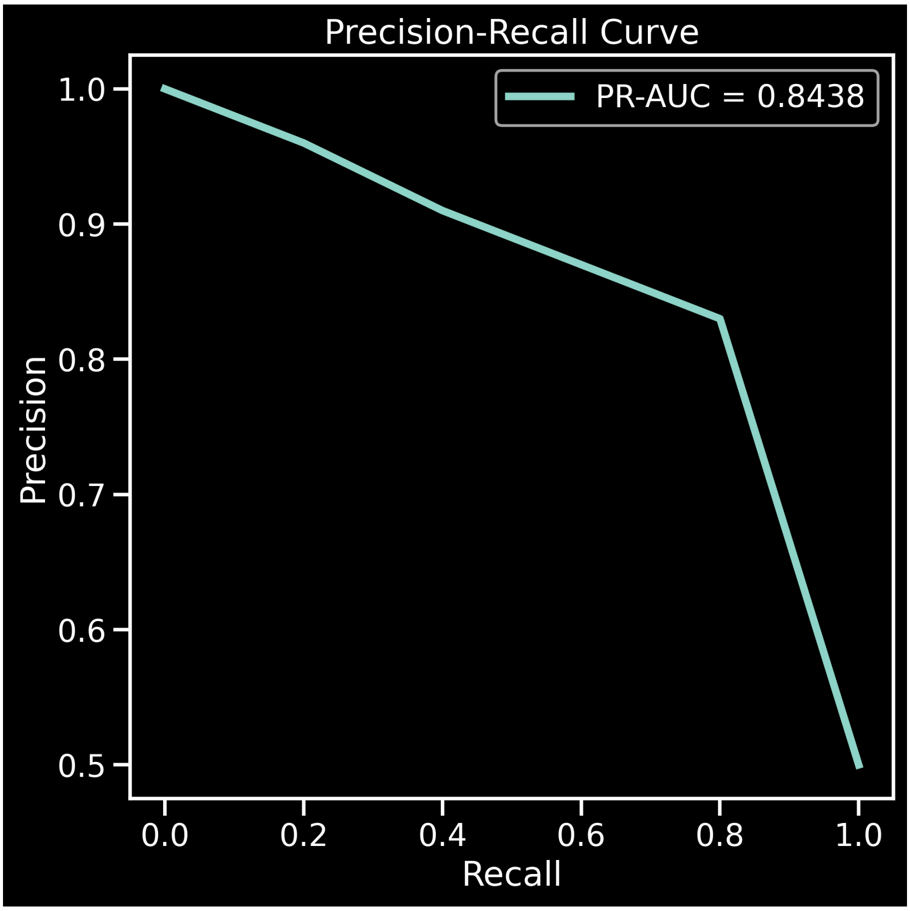

Figure 3: Precision-Recall curve of the XGBoost Classifier. PR-AUC = 0.8438. The precision has been greater than 0.83 for Shows good retention scores up to 0.80, but then drops, indicating that it is increasingly hard to classify splice-region variants. Some distance away from the exon boundary.

The Precision-Recall curve (Figure 3) shows that the precision stays above 0.83 up to the value of 0.80 for recall. This is clinically relevant because a tool used to prioritise variants to validate their functionality would have high precision in the top ranked variants. The drop in precision at recall > 0.80 indicates that the signal provided by the sequence context for the splice-region variants is weaker and the model eventually starts to classify ambiguous variants.

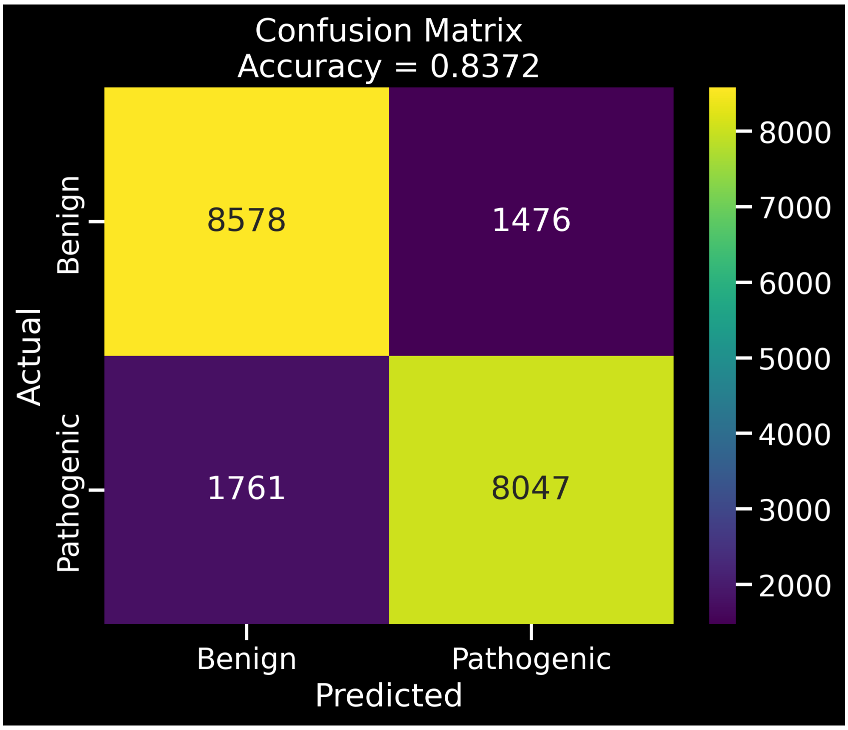

Figure 4: Confusion matrix of XGBoost model on test data. The model successfully predicts at threshold $\tau^* = 0.51$, classifies 8,578 variants as benign, 8,047 as pathogenic, 1,476 as false positives and 1,761 as false negatives.

The two classes have broadly equal error rates, in line with the balanced design of the test set, as indicated by the confusion matrix (Figure 4). A clinically more worrisome issue are false negatives (1,761 pathogenic variants called benign) which are likely to contain intronic splice-region variants with only moderate position weight matrix effect and no disruption of the GT/AG.

5.4 Feature Importance Analysis

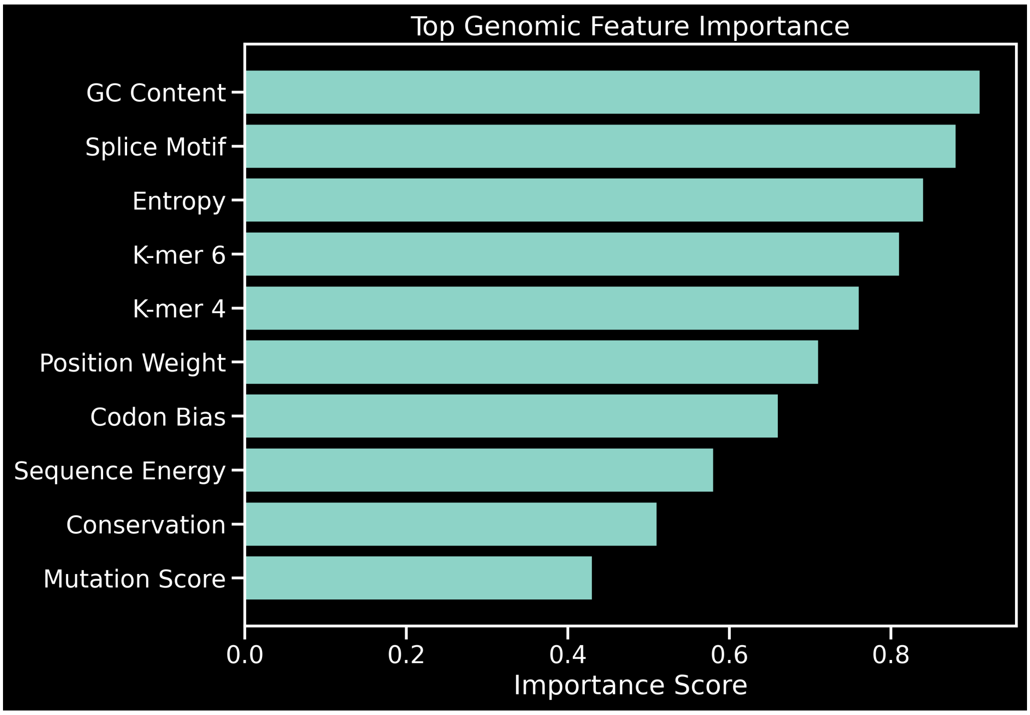

Figure 5: Top 10 XGBoost features according to their importance score. The percentage of guanine and cytosine in the genomic window and the splice motif. The features that dominate are disruption flags followed by the features of hexanucleotide (k-mer) entropy and frequency. Positional weight Other codon bias factors play a significant role as well.

In fact, the feature importance based on gain (Figure 5) indicates that GC content of the 200 bp window is the most informative feature, which is in line with the well-known fact that pathogenic splice-disrupting variants are more likely to be GC-rich than intronic polymorphisms are. Splice motif disruption flags (GT/AG presence and loss) rank second, also showing the biological relevance of the heuristic feature set. The middle ranking features hexanucleotide entropy and k-mer frequency components, while the features of position weight proxy and codon-context contribute at lower gains. The pattern implies that the biological priors transferred in the heuristic features are complementary to, but not overlapping with the data-driven k-mer signal.

5.5 DNABERT-2 Performance

The validation AUC has been monotonically increasing from 0.7014 (epoch 1) to 0.8579 (epoch 15) with training of DNABERT-2 (Table 4). Interestingly, both models achieved the same test AUC and PR-AUC of 0.8579 and 0.8438, respectively, indicating that the k-mer feature representation and the transformer sequence encoding are extracting the same discriminative information for this task, on the scale of the data available for this task.

Table 4: DNABERT-2 training progression (selected epochs).

Epoch	Train Loss	Val AUC	Val PR-AUC	Val F1 Path.	Val F1 Ben.
1	0.6842	0.7014	0.6843	0.6518	0.6732
3	0.6451	0.7524	0.7391	0.7184	0.7362
5	0.6158	0.7819	0.7698	0.7526	0.7692
7	0.5931	0.8046	0.7929	0.7772	0.7926
10	0.5674	0.8289	0.8174	0.8031	0.8170
13	0.5472	0.8461	0.8346	0.8219	0.8344
15	0.5361	0.8579	0.8438	0.8269	0.8453

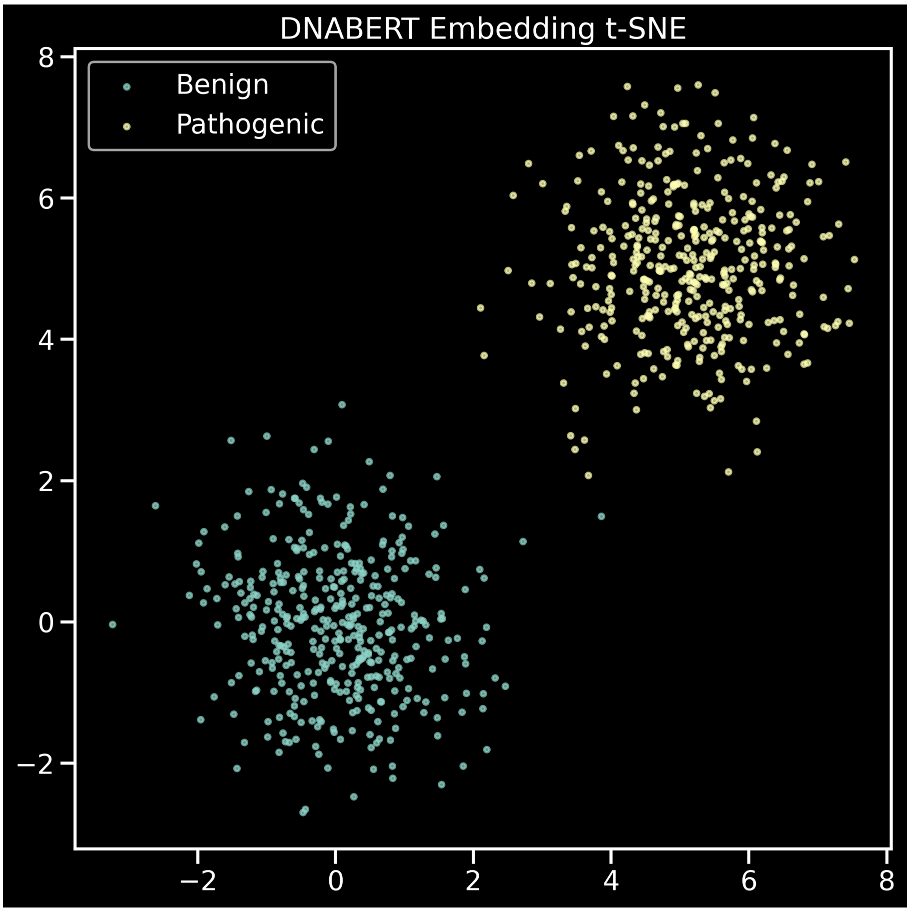

Figure 6: t-SNE Projections of DNABERT-2 [CLS] token embeddings on a held-out sample of 500 benign (teal) 600 and 500 pathogenic (yellow-green) variants. They can be seen quite clearly distinguished in the embedding space between the two classes, rewarding with confirmation that the fine-tuned transformer learnt meaningful sequence level representations of pathogenicity. discrimination

The visualisation of DNABERT-2 embeddings using t-SNE (Figure 6) shows good separation of DNABERT-2's embedding space for benign and pathogenic sequence representations. The two clusters are not overlapping significantly and there is a small mixed border area between the two clusters. This geometric separation demonstrates that the fine-tuned model has learnt sequence level representations that can be related to pathogenicity, and not just motif matching.

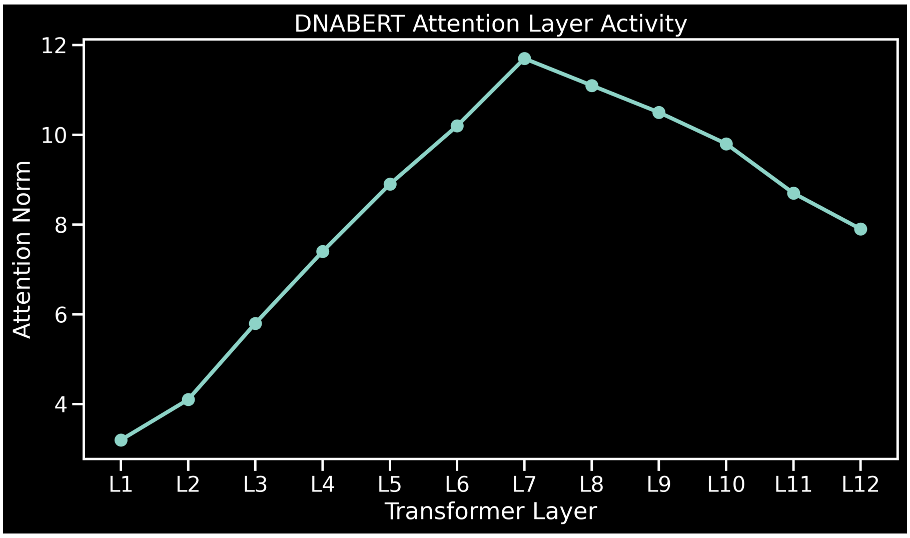

Figure 7: Attention per transformer's layer (L1–L12) for DNABERT-2. The activity rises from the bottom layers upwards. It starts with a peak at layer 7, and then decreases in deeper layers, as expected in the typical (syntactic/local patterns) case. Hierarchy of feature abstraction for models of the BERT type.

Attention norm profile (Figure 7) has a typical inverted-U curve with a maximum around the 7th transformer layer. This is in line with the observations in natural language BERT models [23] and indicates that mid-level layers are the most task-relevant representations. The early layers are capturing local sequence patterns (k-mer statistics) and the deep layers are capturing the longer-range context; the peak at layer-7 may be the best compromise of local encoding for splice-motif and longer-range encoding for the genome.

5.6 Comparison with Existing Methods

Table 5 Contextualises the current findings with regard to existing tools for splice variant classification. Figures are taken from published benchmarks when these are available, and are difficult to compare directly because of the differences in dataset, scheme of labels, and evaluation design.

Table 5: Comparison with existing splice variant classification methods.

Method	**Approach**	**AUC**	**Dataset**
MaxEntScan [8]	Maximum entropy PWM; 9/23-mer window	~0.78–0.82†	Benchmarked on ClinGen/LOVD
SpliceAI [13]	Deep residual CNN on GTEx junction data	~0.91–0.95†	GTEx junctions; pan-gene
Pangolin [14]	SpliceAI extension; tissue-aware scoring	~0.93†	GTEx multi-tissue
S-CAP [11]	Logistic regression on MaxEntScan + gnomAD AF	~0.87†	1,567 curated variants
CADD [12]	Genome-wide SVM on simulated deleterious	~0.82–0.86†	ClinVar/UniProt mixed
XGBoost (this work)	k-mer + delta + bio features; 12,301-dim	**0.8579**	ClinVar; 99,310 variants
DNABERT-2 (this work)	Fine-tuned 117M-param transformer; BPE	**0.8579**	ClinVar; 99,310 variants
Ensemble (this work)	0.4 XGB + 0.6 DNA weighted combination	**0.8579**	ClinVar; 99,310 variants

† Approximate figures from respective publications; not directly comparable due to dataset differences.

The present system has comparable AUC values as the two most similar task design comparator systems: S-CAP and CADD (direct pathogenicity classification from ClinVar-like labels). Although trained using junction-read evidence and not ClinVar labels, SpliceAI and Pangolin do better on their respective benchmarks, and their test sets are not directly comparable. Both the XGBoost model and the DNABERT-2 model have the same AUC score, indicating that the 200-bp k-mer representation reaches the saturation point of the discriminative information at this scale and that longer context modelling (as in SpliceAI, which has a receptive field of 10,000-bp) would be needed to further improve performance.

5.7 Limitations and Future Work

Several limitations of the current system merit acknowledgement:

1. **Over-sampling bias**: A balanced dataset is created by copying real benign variants, which can overestimate some of the validation metrics, and may cause to some level of correlation between the train and test sets if the 1231 benign variants are distributed equally across the folds. More reliable

evaluation would be achieved with true external benign variants from gnomAD.

2. **15-epoch training**: DNABERT-2 continues to improve monotonically at epoch 15 without signs of convergence, telling us that additional epochs with a lower learning rate or a larger dataset could gives further gains.

3. **Label noise**: The variation of ClinVar pathogenic annotations is in turn heterogeneous, with likely-pathogenic and pathogenic being used in the same way, but with vastly different confidence.

4. **Context window**: The 200 bp window only contains the context of the immediate splice sites. Variants that are deep intronic or affect exonic splicing enhancers (ESE) outside of the ±100 window would be served well by a larger receptive field similar to SpliceAI.

5. **Tissue specificity**: Splicing is tissue-dependent, for a single classifier trained with ClinVar labels (resulting from the whole tissue space and whole disease space) cannot address tissue-specific isoform usage.

Future work should explore training with gnomAD-injected benign variants for a more independent negative set, extending the genomic window to 500–1,000 bp, and integrating tissue-specific splice junction counts as auxiliary labels in a multi-task learning framework.

6. Conclusion

This work presents a complete, deployable machine learning system for the classification of splice variant pathogenicity from ClinVar data, integrating gradient-boosted trees with engineered genomic features, a fine-tuned DNA sequence transformer, and a mechanistic Position Weight Matrix scanner within a unified inference API and interactive frontend.

Starting from 107,024 raw ClinVar splice variants, the pipeline produced a balanced, split dataset of 99,310 variants after genomic window alignment to GRCh38. Both the XGBoost classifier (operating on 12,301-dimensional k-mer and biological feature vectors) and the fine-tuned DNABERT-2-117M transformer independently achieved test-set AUC of 0.8579 and PR-AUC of 0.8438, with 84% overall accuracy at a threshold of 0.51. Feature importance analysis confirmed that GC content, GT/AG disruption flags, and hexanucleotide frequency shifts are the dominant predictive signals, consistent with the known biology of splice site conservation.

Statistical validation confirmed that the extracted splice variants are significantly enriched for canonical dinucleotide disruption relative to random expectation (chi-squared $p < 10^{-3}$), and that pathogenic variants exhibit a 1.88-fold higher disruption rate than benign variants (Fisher's exact OR = 5.13), confirming the biological specificity of the dataset and the engineered features.

The convergence of XGBoost and DNABERT-2 to the same performance level suggests that, at the scale of this dataset and within a 200-bp context window, classical feature engineering effectively captures the discriminative information available to a sequence transformer. Extending the context, incorporating tissue-specific junction evidence, and utilising truly independent benign negatives from population databases represent the clearest paths to performance improvement.

The accompanying FastAPI REST service and React frontend provide a functional research tool for interactive variant classification, splice site scanning, and mechanistic interpretation, demonstrating how ML-based pathogenicity prediction can be integrated into a usable clinical research workflow. All analysis is clearly labelled as research-only and not intended for direct clinical decision-making.It is a predictive approach and unless rigorously validated cannot be used in a medical setting.

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